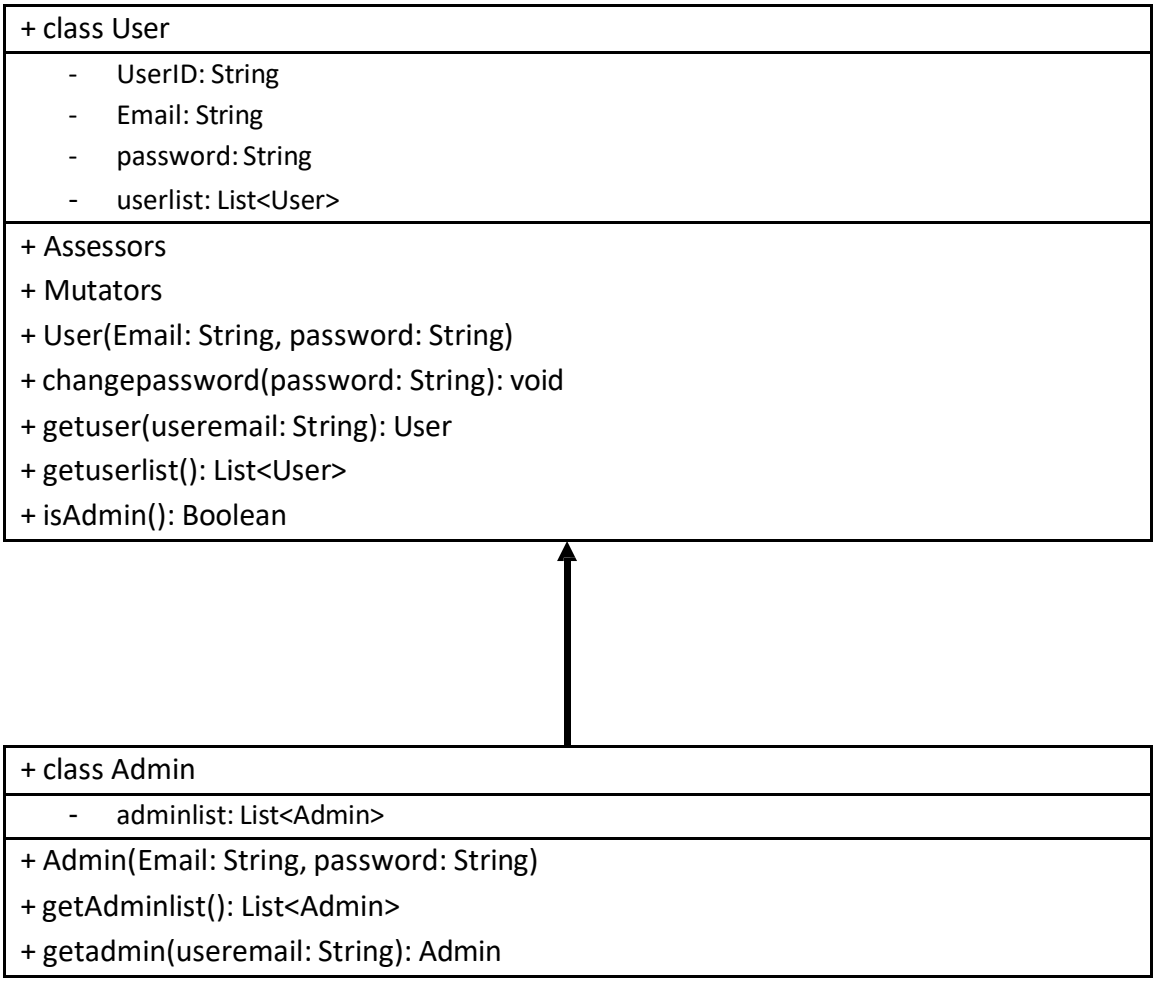
Criterion B: Designed Overview

UML Diagram for Class User:



Overview of Program:

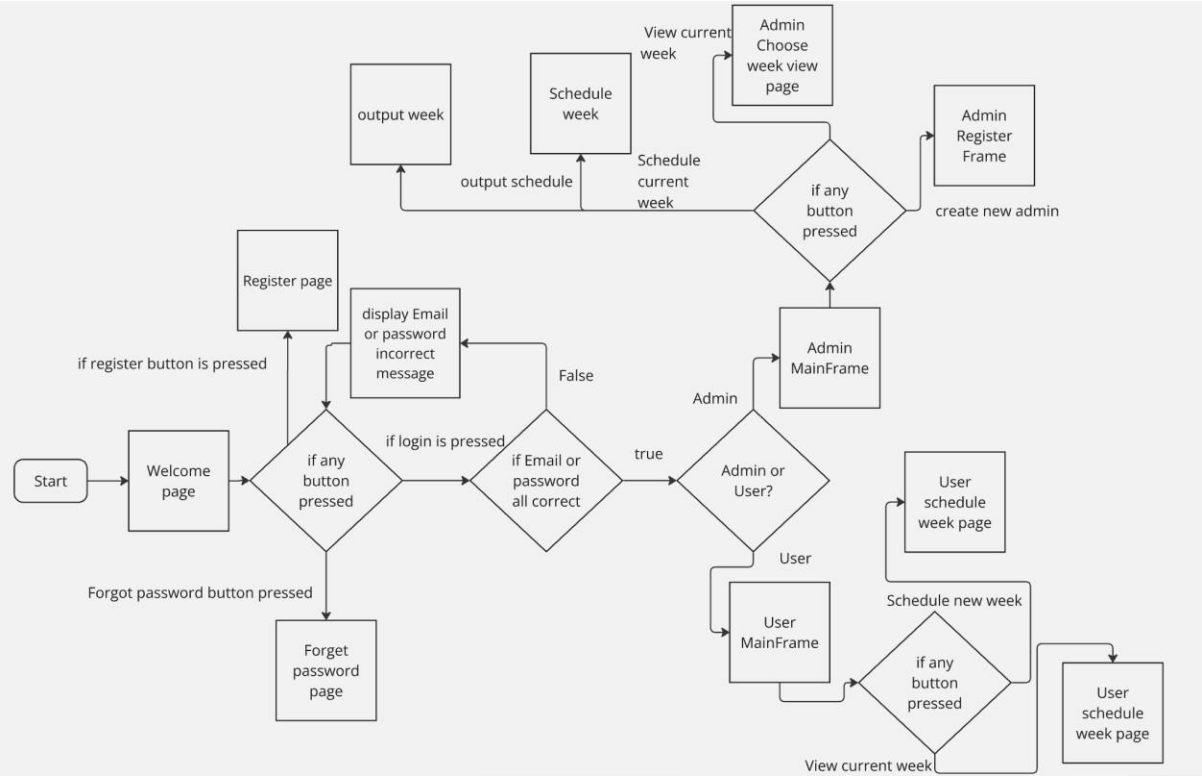


Figure 1 Overview of program

1. Login System

1.1 Welcome Page:

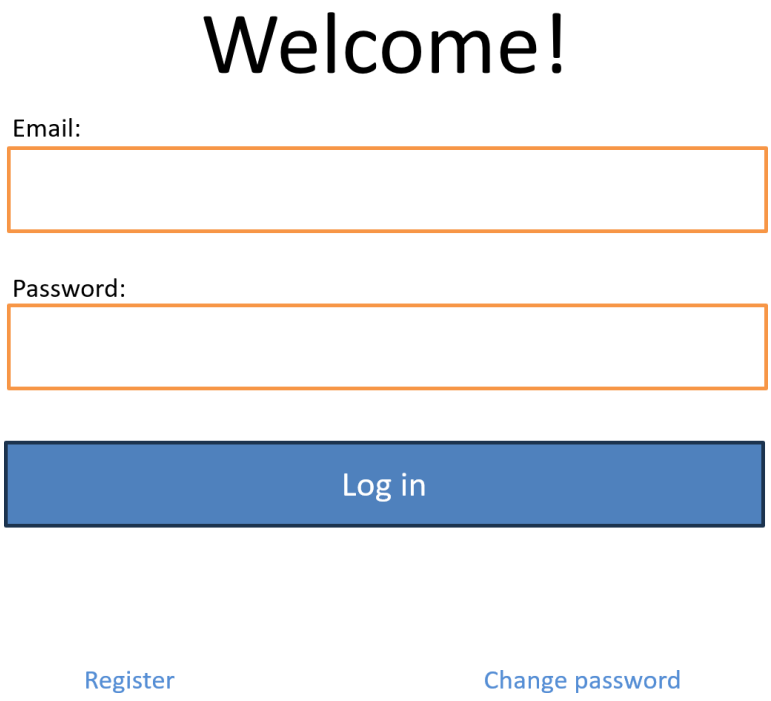


Figure 2.0 Welcome interface

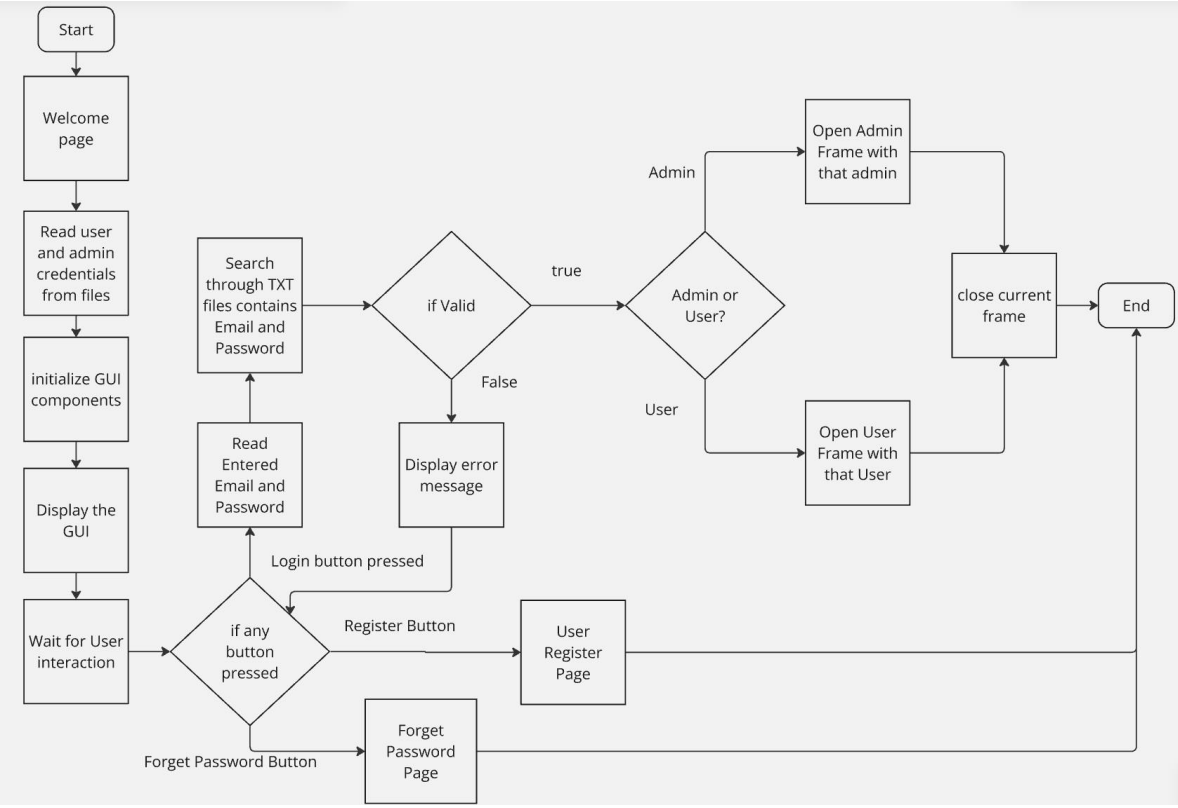


Figure 2.1 Welcome Page Flow chart

+ class Welcome – extends JFrame – implements ActionListener
- Emaildocumentpath: String - Passworddocumentpath: String - AdminEmaildocumentpath: String - AdminPassworddocumentpath: String - SouthPanel: JPanel - Loginbutton: JPanel - textfield: JPanel - buttons: JPanel - Welcome: JLabel - Emaillabel: JLabel - Passwordlabel: JLabel - Email: JTextField - PasswordEnter: JPasswordField - ForgotPassword: JButton - Register: JButton - Login: JButton - INPUT: String - Password: String
+ Welcome() + actionPerformed(e: ActionEvent) - getList(path: String): List - check(emailPath: String, passwordPath: String, passwordEntered: String,

emailEntered: String): boolean

Figure 2.2 Welcome Page Class UML diagram

1.2 User Register Page

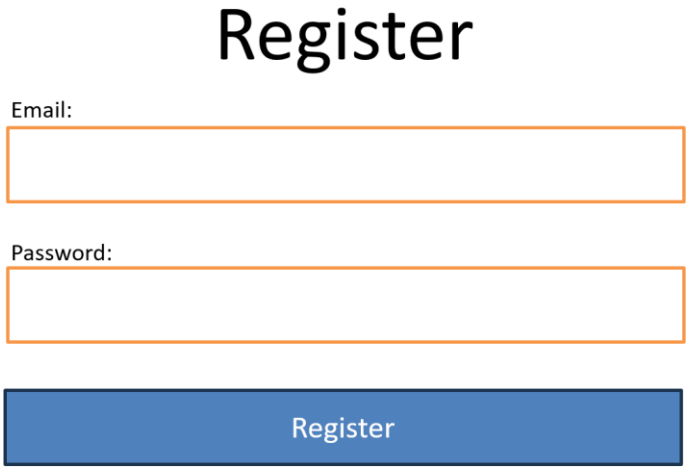


Figure 3.0 Register Page Diagram

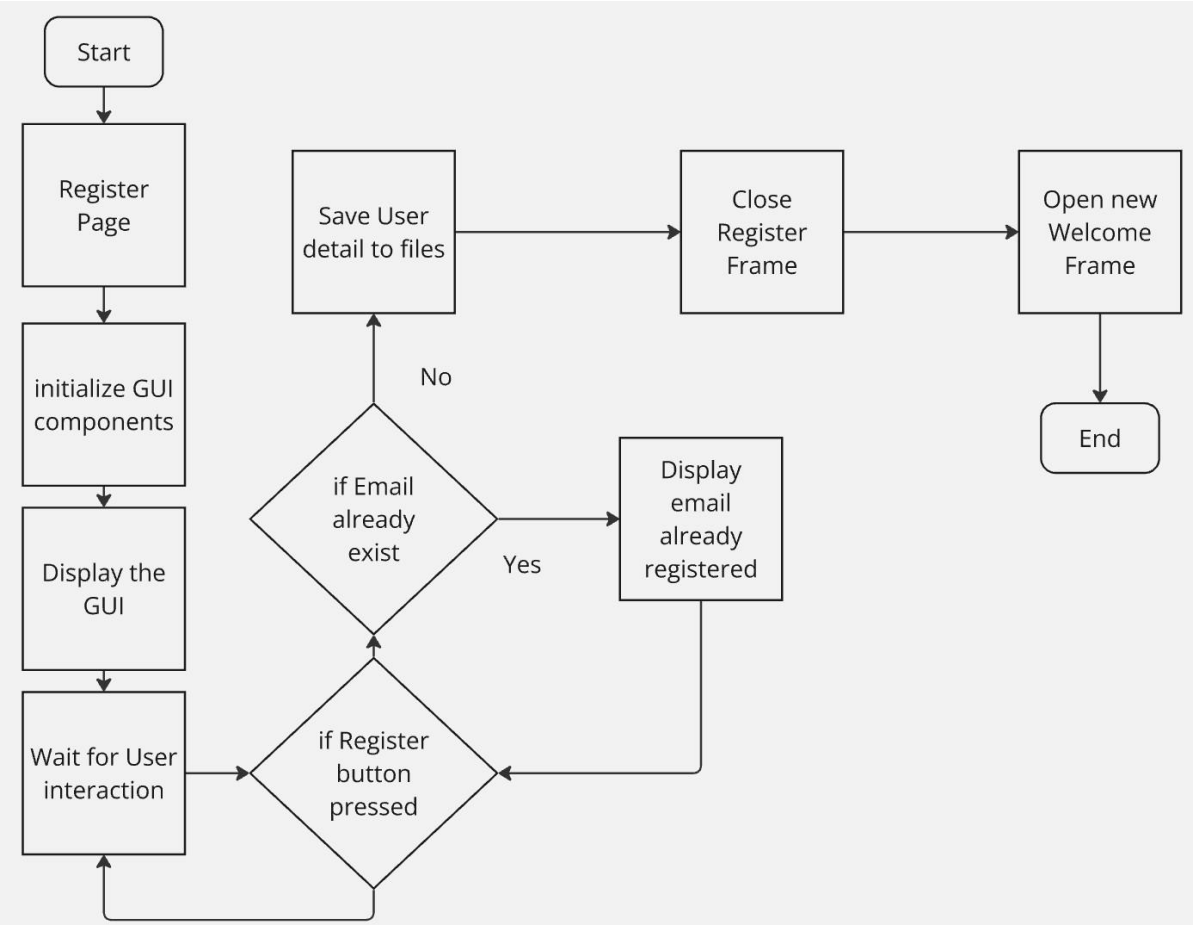


Figure 3.1 Register Page Flow Chart

+ class RegisterPage – extends JFrame – implements ActionListener
- Emaildocumentpath: String - Passworddocumentpath: String - Register: JLabel - Emaillabel: JLabel - codelabel: JLabel - Registerbutton: JButton - Email: JTextField - PasswordEnter: JPasswordField - textfieldpanel: JPanel - SouthPanel: JPanel
+ RegisterPage() + actionPerformed(e: ActionEvent) : void - IsEmailRegistered(email: String) : boolean

Figure 3.2 Register Page UML Diagram

1.3 Change Password Page:

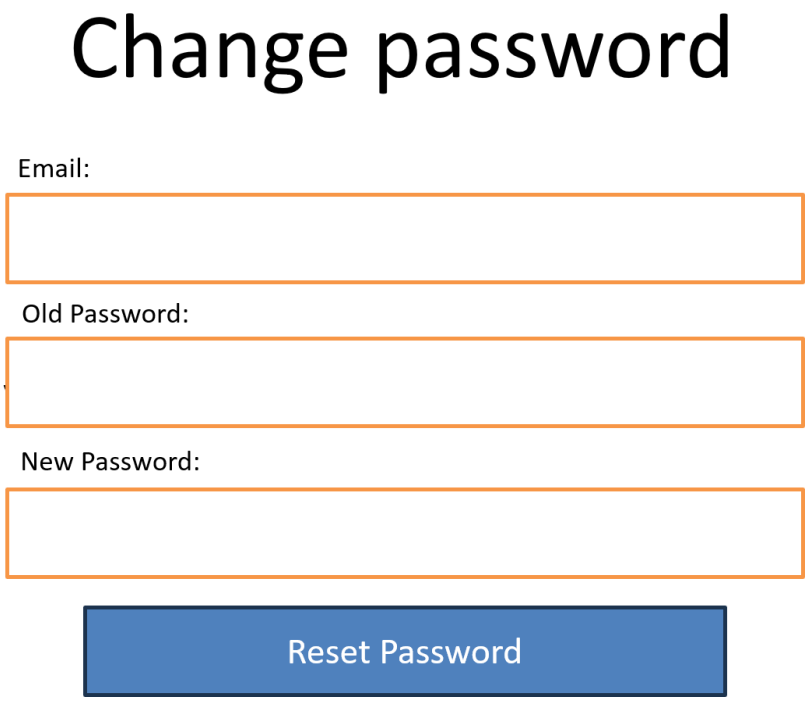


figure 4.0 Change Password Interface

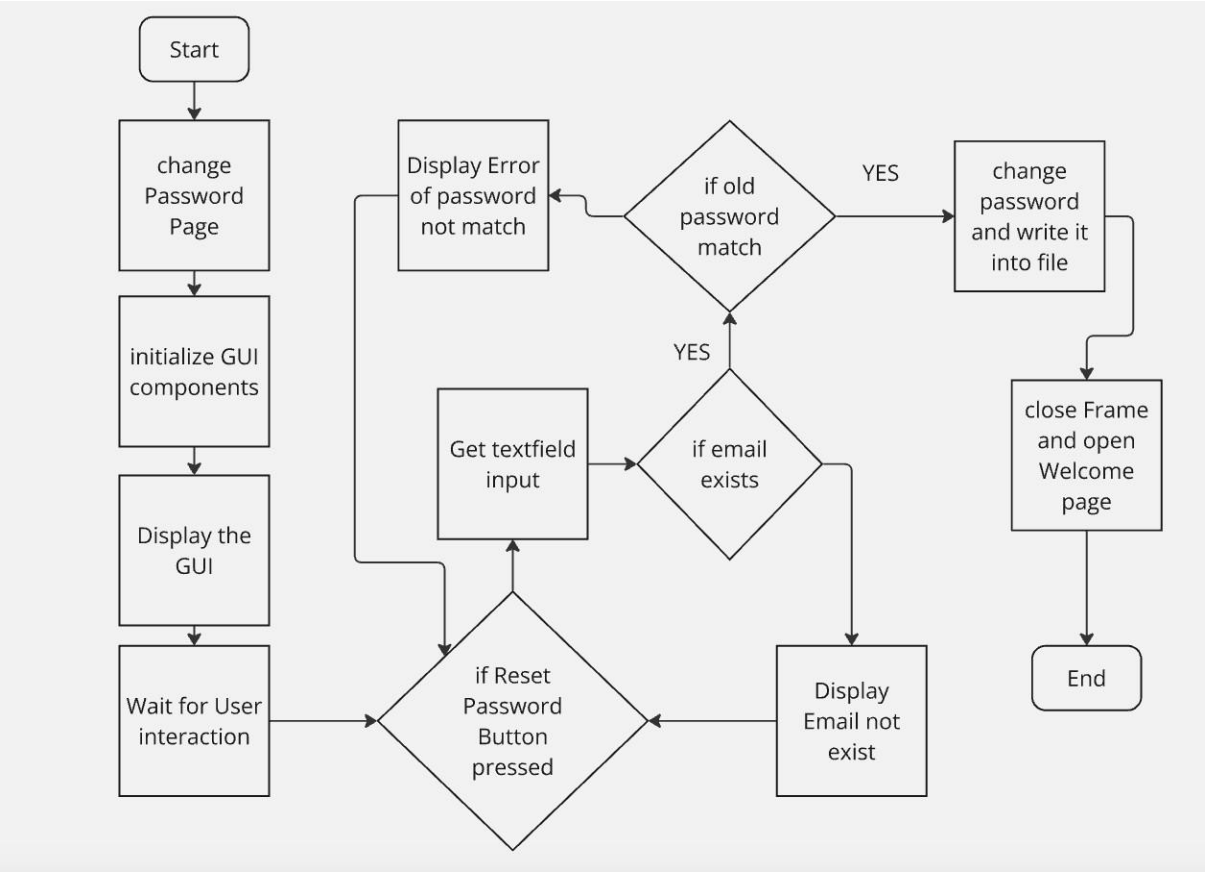


Figure 4.1 Change Password Page Flow chart

+ class ChangePasswordPage – extends JFrame – implements ActionListener
- Emaildocumentpath: String - Passworddocumentpath: String - ForgetPassword: JLabel - Emaillabel: JLabel - oldPasswordLabel: JLabel - newPasswordLabel: JLabel - resetpasswordbutton: JButton - Email: JTextField - oldPasswordField: JPasswordField - newPasswordField: JPasswordField - textfieldpanel: JPanel - SouthPanel: JPanel
+ ChangePasswordPage() + actionPerformed(e: ActionEvent) : void - GetList(path: String): List<String>

Figure 4.2 ChangePasswordPage UML Diagram

2 User Frame design

2.0 User Main Frame

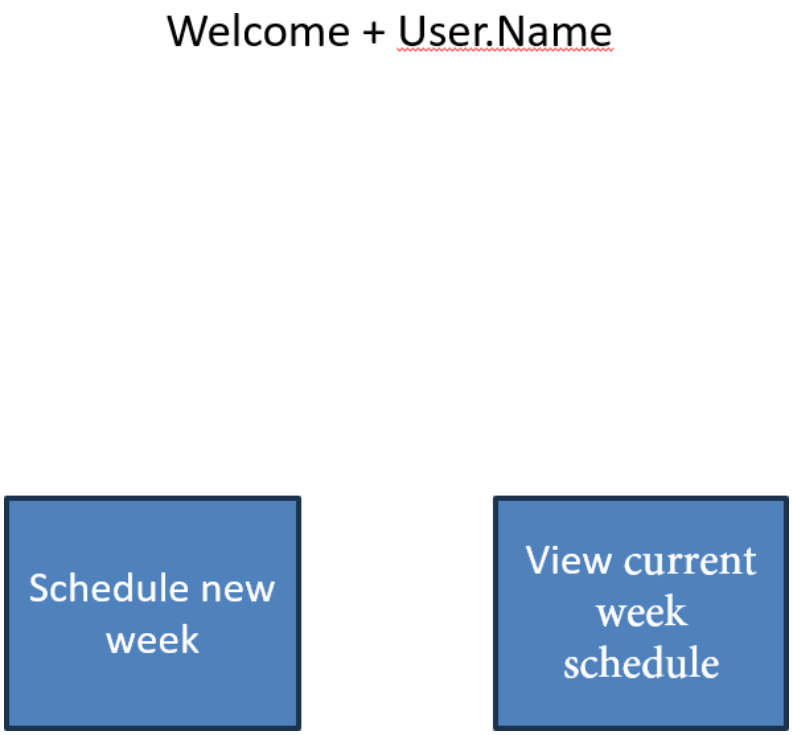


Figure 5.0 MainFrame for User

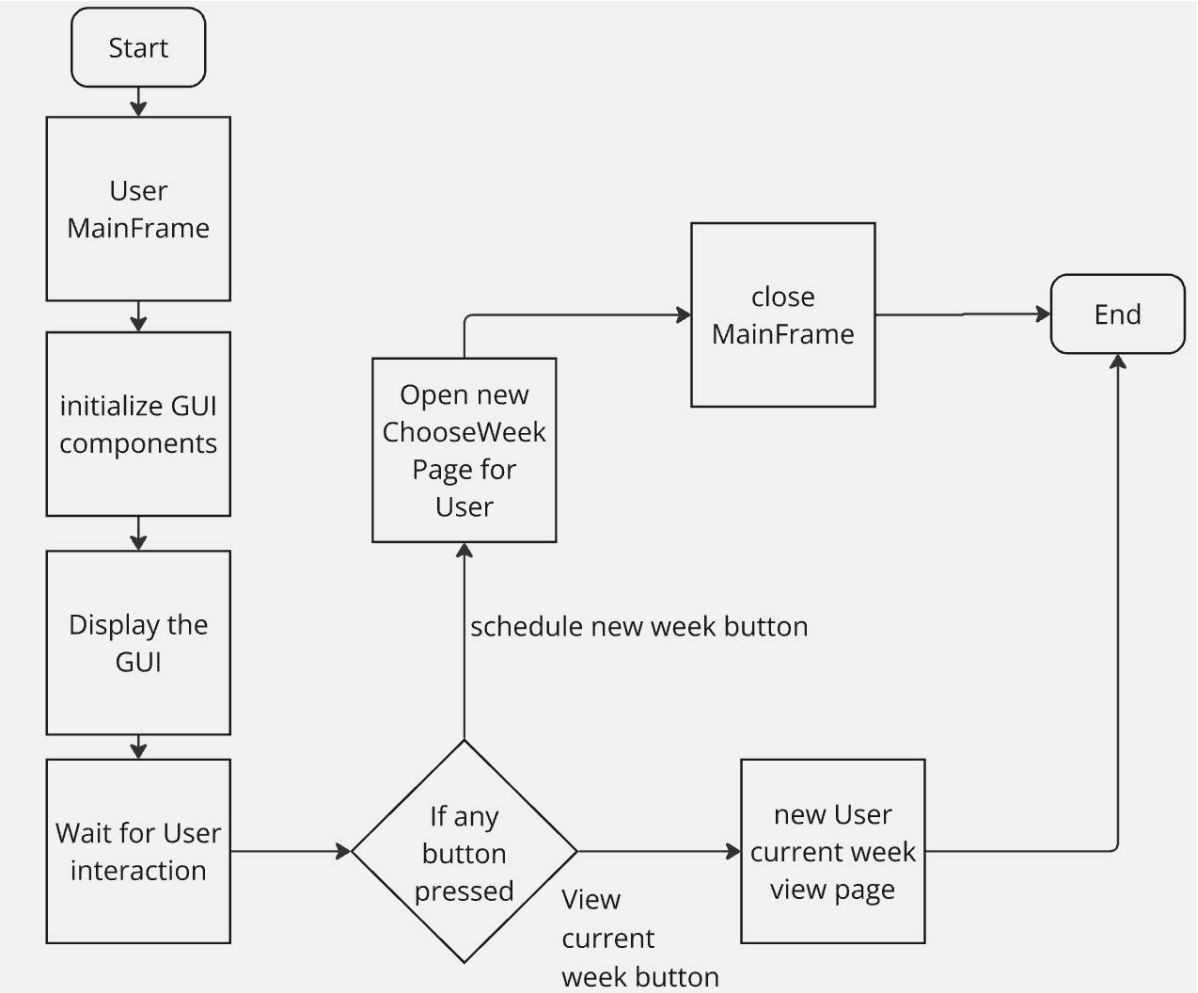


Figure 5.1 User MainFrame Flowchart

+ class MainFrame– extends JFrame – implements ActionListener
- user: User - Schedule_new_week: JButton - View_current_week_schedule: JButton - Personal_detail: JButton - label: JLabel - buttonpanel: JPanel - southPanel: JPanel
+ MainFrame(user : User) + actionPerformed(e: ActionEvent) : void

Figure 5.2 User MainFrame UML Diagram

2.1 User Choose Week Page:

<- Back

Schedule week

Date	Week number	A / B	
			-> Schedule
			-> Schedule
			-> Schedule
			-> Schedule

Figure 6.0 User ChooseWeekPage interface

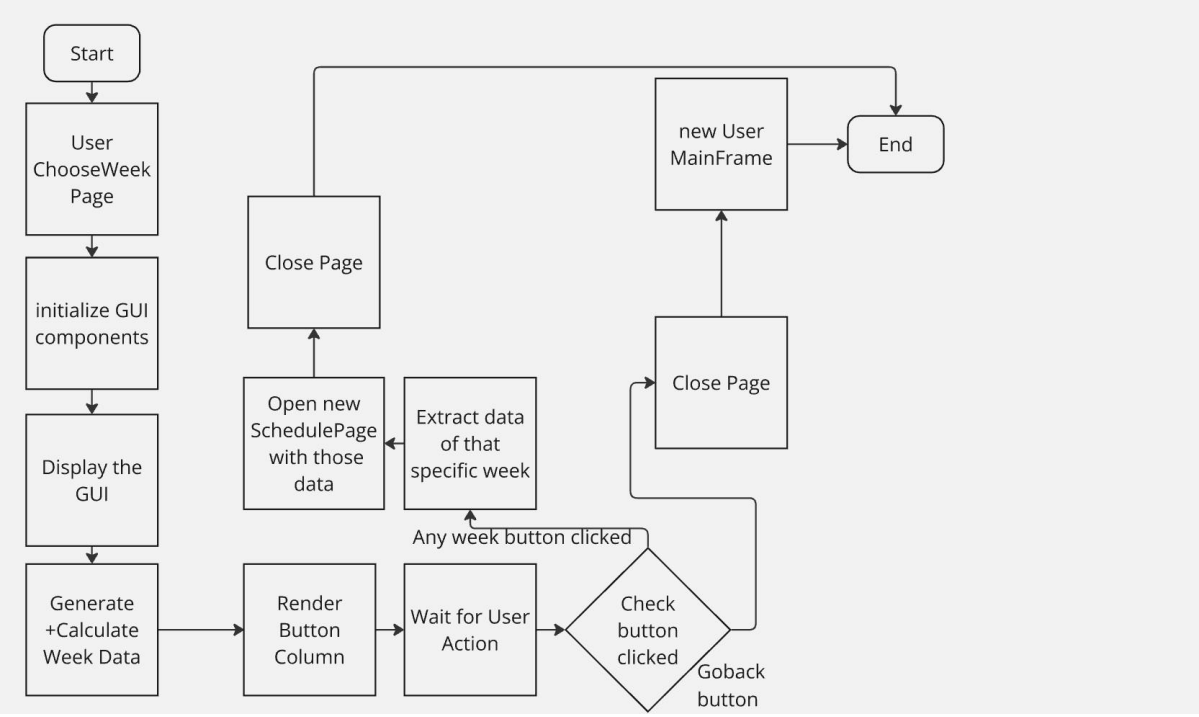


Figure 6.1 Flowchart for User ChooseWeekPage

+ class ChooseWeekPage extends JFrame
- user: User - schedule: JTable - tableModel: DefaultTableModel - goback: JButton - scheduleWeek: JLabel
+ Chooseweekpage(user: User) + generateScheduleWeekData(): void

Figure 6.2 ChooseWeekPage class UML diagram

Class contains:

+ class ButtonEditor - extends DefaultCellEditor
- button: JButton - date: String - weekNumber: int - weekAB: String
+ ButtonEditor(checkBox: JCheckBox) + getTableCellEditorComponent(JTable table, Object value, boolean isSelected, int row, int column) : Component + getCellEditorValue(): Object

Figure 6.3 ButtonEditor class UML diagram

Class contains:

+ class ButtonRenderer - extends JButton – implements javax.swing.table.TableCellRenderer
+ ButtonRenderer() + getTableCellRendererCompoment(JTable table, Object value, boolean isSelected, boolean hasFocus, int row, int column) : Component

Figure 6.4 ButtonRenderer class UML Diagram

2.2 User Schedule Table Page:

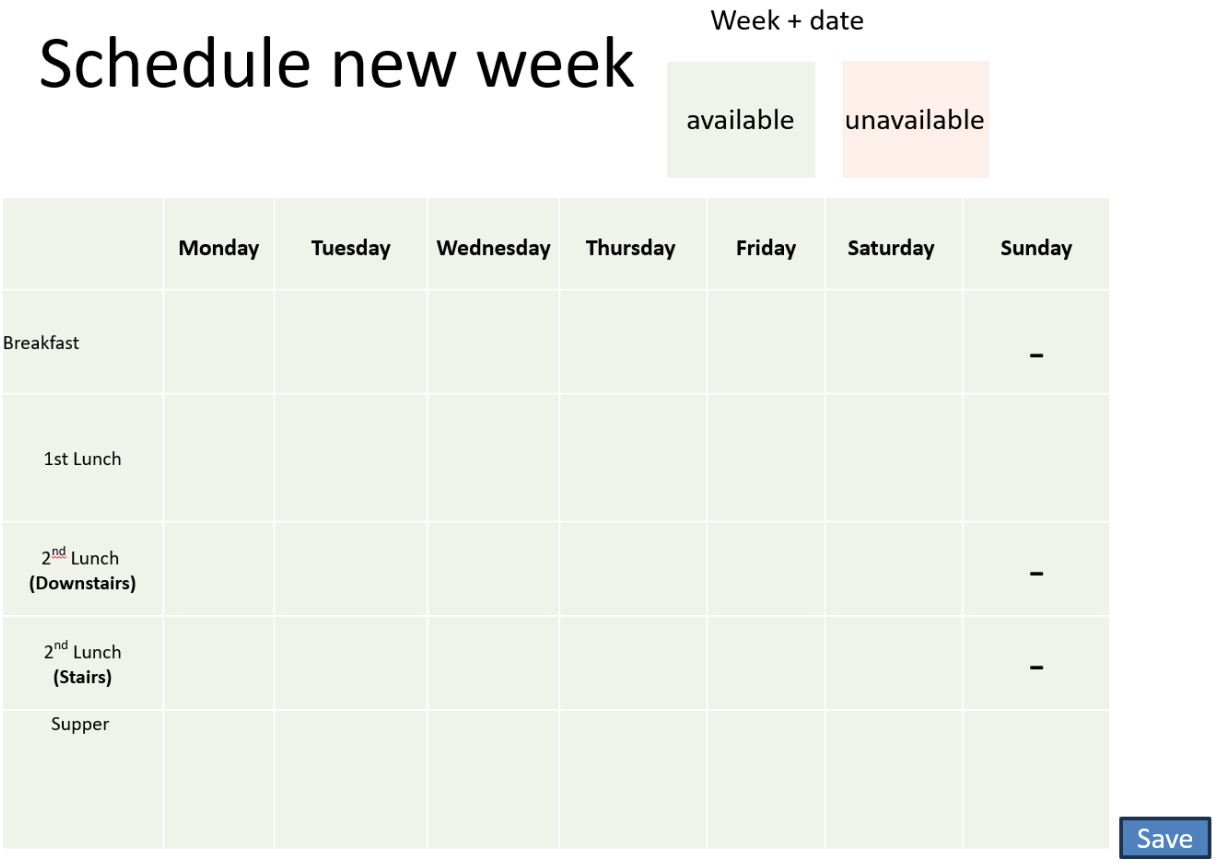


Figure 7.0 ScheduleTablePage interface

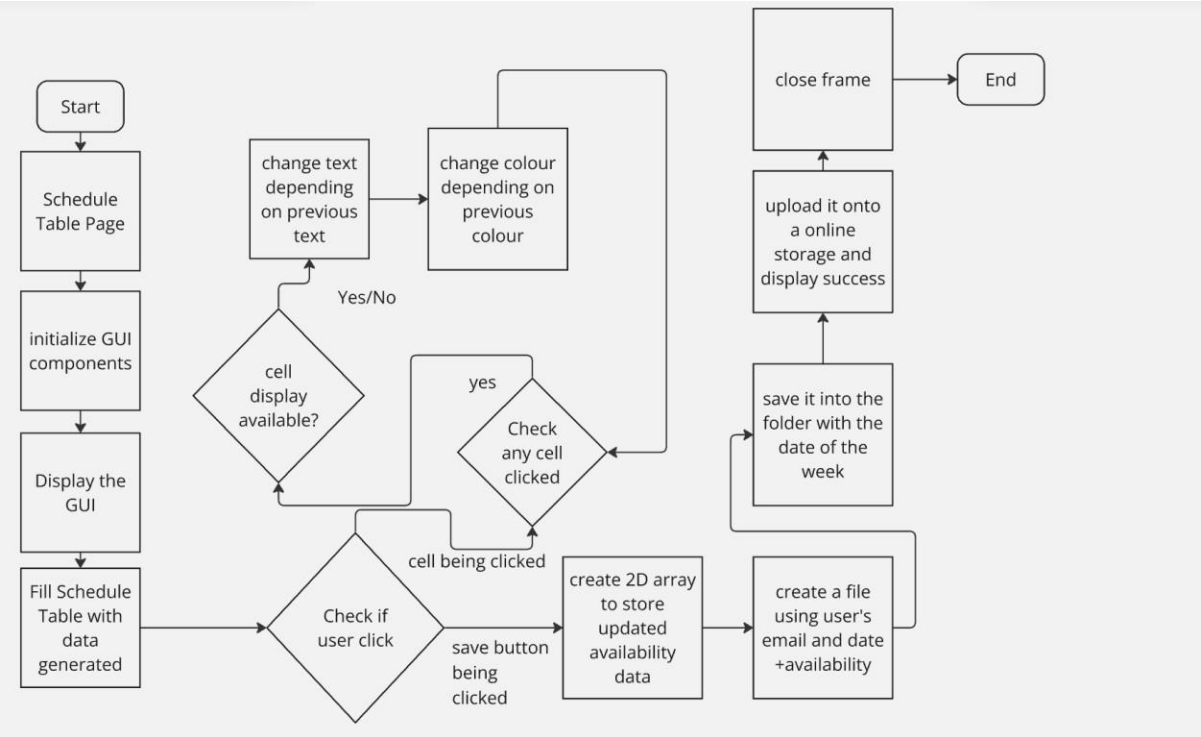


Figure 7.1 ScheduleTablePage flowchart

+ class Scheduletablepage - extends JFrame – implements ActionListener
- scheduleTable: JTable - tableModel: DefaultTableModel - saveButton: JButton - buttonpanel: JPanel - panel: JPanel - timeSlots: String[][] - date: String - weekNumber: int - weekAB: String - titlelabel: JLabel - titlepanel: JPanel - noticelabel: JLabel - noticepanel: JPanel - user: User
+ Scheduletablepage(date, weekNumber, weekAB, user) + actionPerformed(e: ActionEvent)

Figure 7.2 Scheduletablepage class UML Diagram

Class contains:

+ class CustomCellRenderer – extends DefaultTableCellRenderer
+ getTableCellRendererCompoment(JTable table, Object value, boolean isSelected, boolean hasFocus, int row, int column) : Component

Figure 7.3 CustomCellRenderer class UML Diagram

2.3 Choose week to view for User:

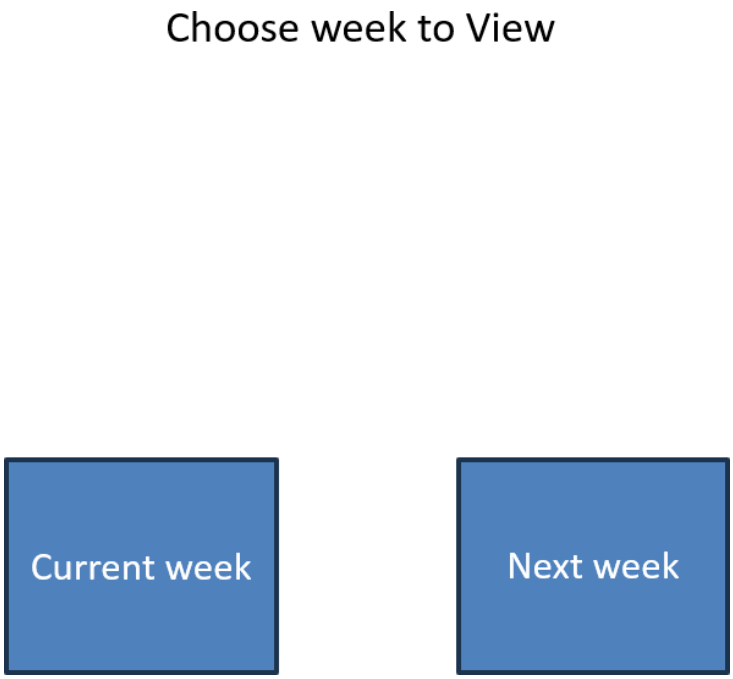


figure 8.0 ChooseWeekViewPage interface

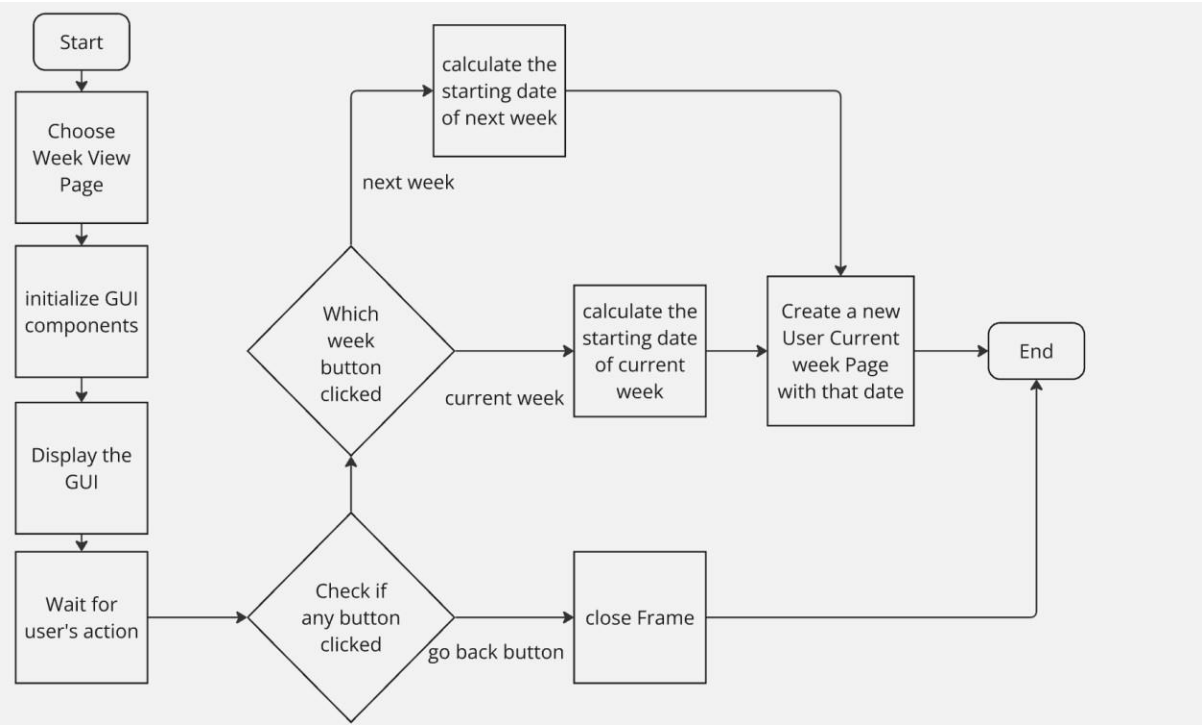


Figure 8.1 ChooseWeekViewPage Flowchart

+ class ChooseWeekViewPage- extends JFrame – implements ActionListener
- user: User - currentWeekButton: JButton - nextWeekButton: JButton - goBackButton: JButton - buttonpanel: JPanel
+ ChooseWeekViewPage(user: User) + actionPerformed(e: ActionEvent)

Figure 8.2 ChooseWeekViewPage UML Diagram

2.4 UserCurrentweekpage – display week schedule:

Current week schedule

Date + week

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Breakfast							-
1st Lunch							
2nd Lunch **(Downstairs)**							-
2nd Lunch **(Stair)**							-
Supper							

Figure 9.0 UserCurrentWeekPage interface

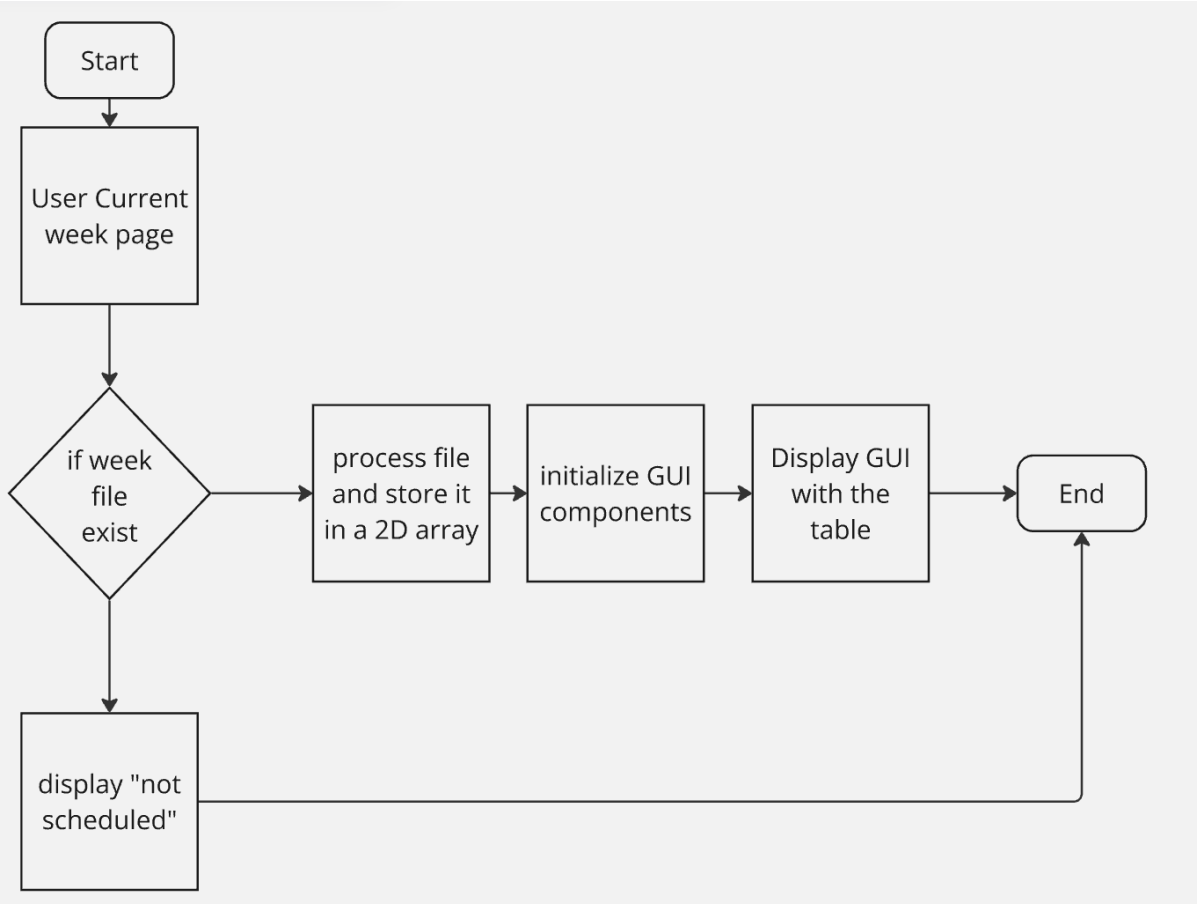


Figure 9.1 UserCurrentWeekPage flowchart

+ class UserCurrnetWeekPage- extends JFrame
- weekNumber: int - weekAB: String - titlelabel: JLabel - titlepanel: JPanel - table: JTable - tablemodel: DefaultTableModel - user: User - date: String
+ UserCurrentweekpage(date: String, user: User) + processFile(file: File): String[][]

Figure 9.2 UserCurrentWeekPage class UML Diagram

3 Admin Frame design

3.0 AdminMainFrame:

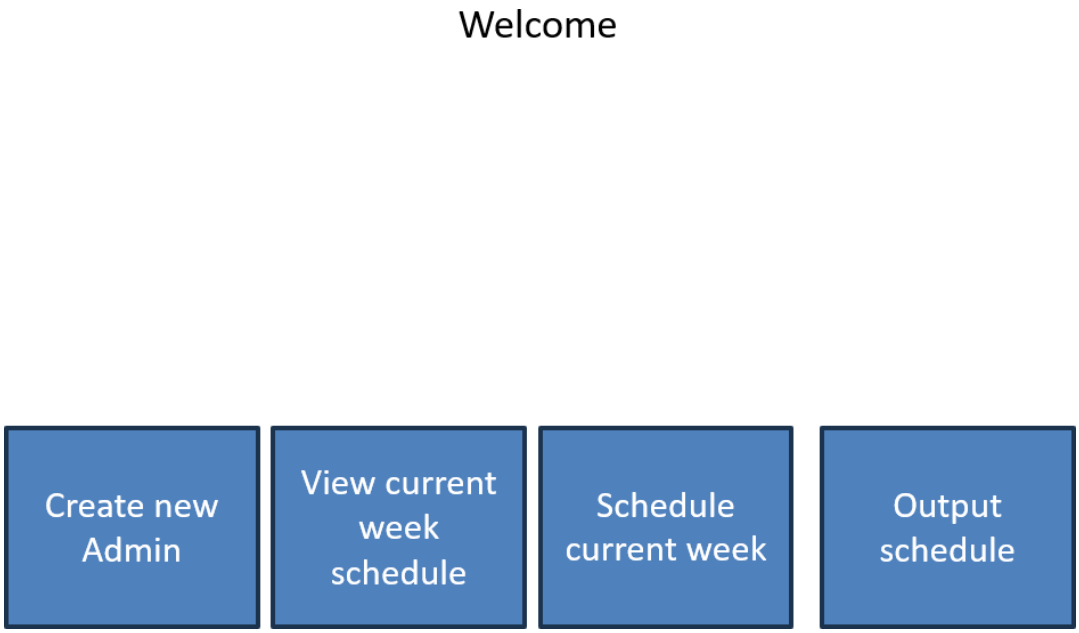


Figure 10.0 AdminMainFrame interface

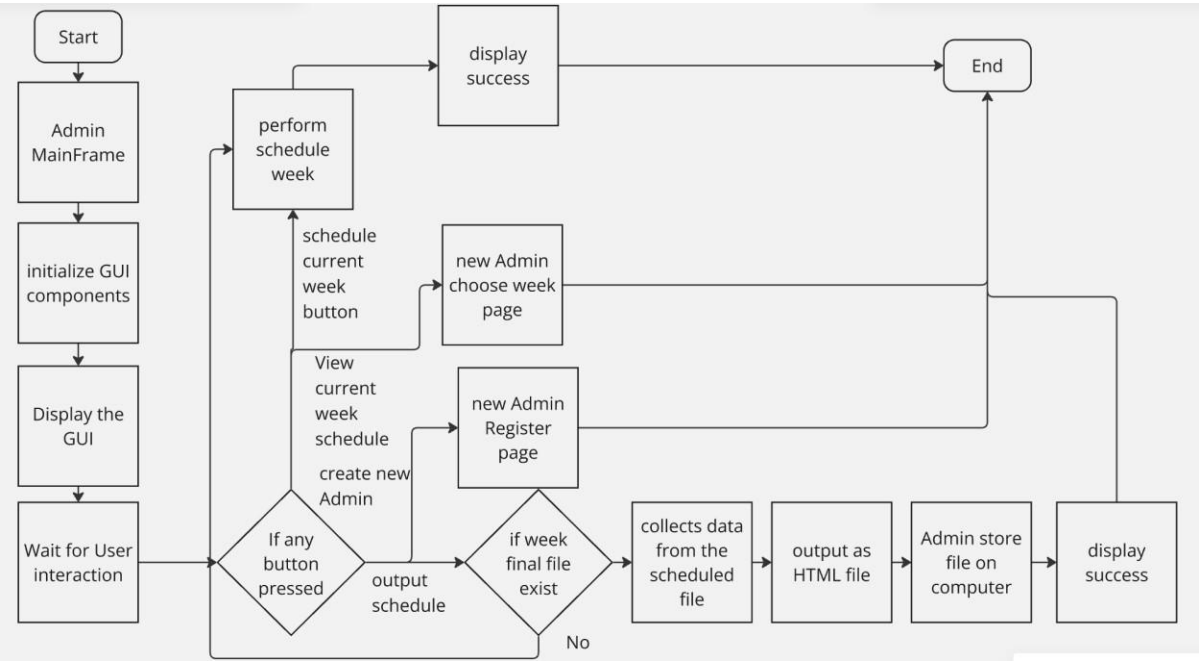


Figure 10.1 AdminMainFrame flowchart

+ class AdminMainFrame - extends JFrame – implements ActionListener
- admin: Admin - label: JLabel - buttonpanel: JPanel - southPanel: JPanel - create_new_Admin: JButton - View_current_week_schedule: JButton - Schedule_current_week: JButton - Schedule_to_picture: JButton - COLUMN_HEADERS: String[] - ROW_HEADERS: String[]
+ AdminMainFrame(Admin admin) + actionPerformed(e: ActionEvent): void - schedulefinalarray(startOfWeekDate: String): String[][] - extractEmailFromFileName(fileName: String): String - processFile(file: File): String[][] - organizeDataForLocations(emailToDataMap: Map<String, String[][]>): String[][]

Figure 10.2 AdminMainFrame class UML Diagram

3.1 Admin Choose Week Page:

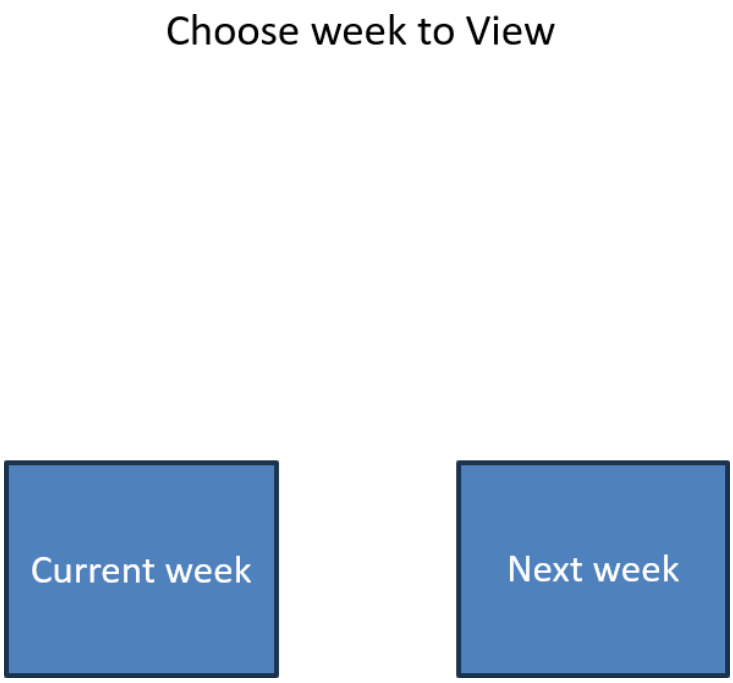


Figure 11.0 - Same as User – Choose Week View Page interface

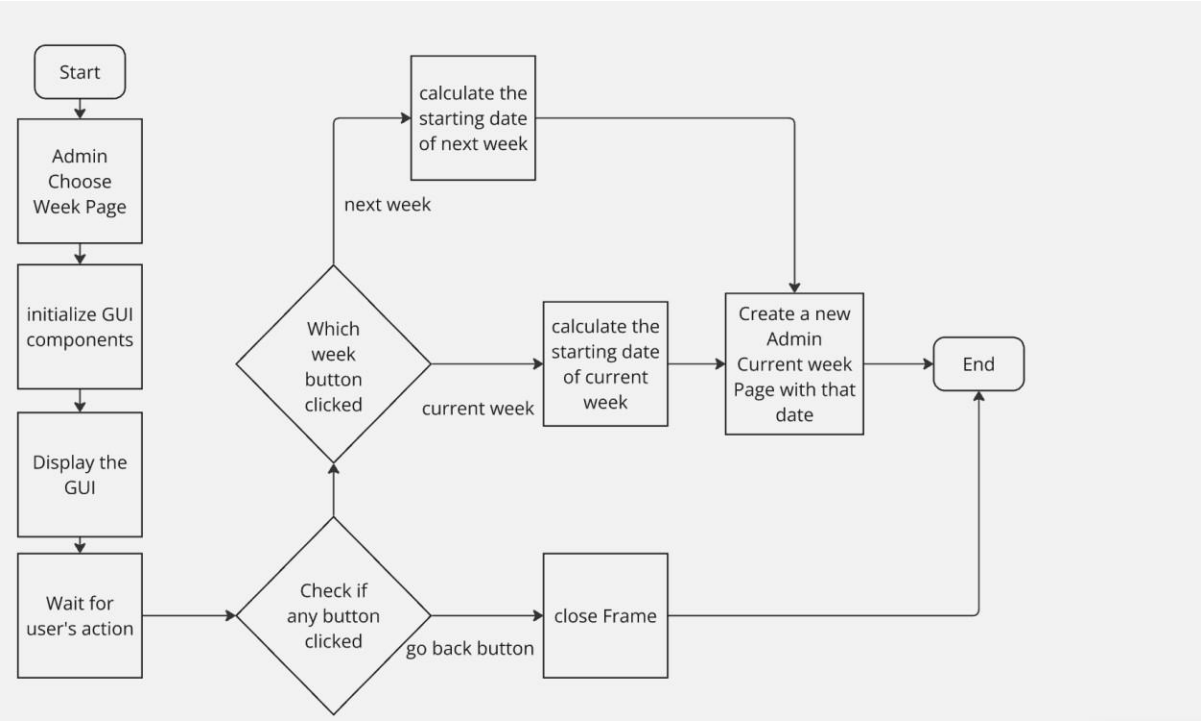


Figure 11.1 - Admin Choose Week Page flowchart

+ class AdminMainFrame - extends JFrame – implements ActionListener
- admin: Admin - currentWeekButton: JButton - nextWeekButton: JButton - goBackButton: JButton - buttonpanel: JPanel
+ AdminChooseWeekViewPage(admin: Admin) + actionPerformed(e: ActionEvent): void

Figure 11.2 AdminChooseWeekPage class UML Diagram

3.2 Admin Register Page:

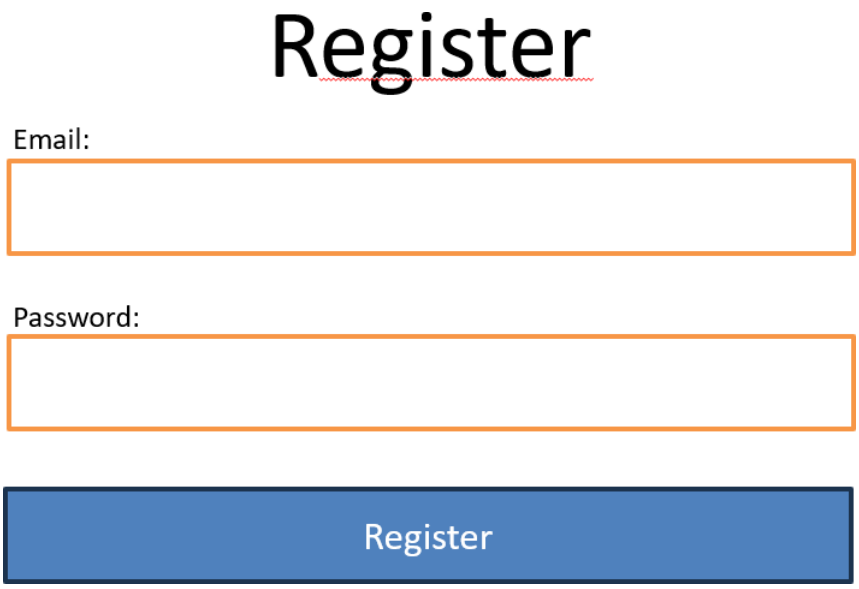


Figure 12.0 AdminRegisterPage interface

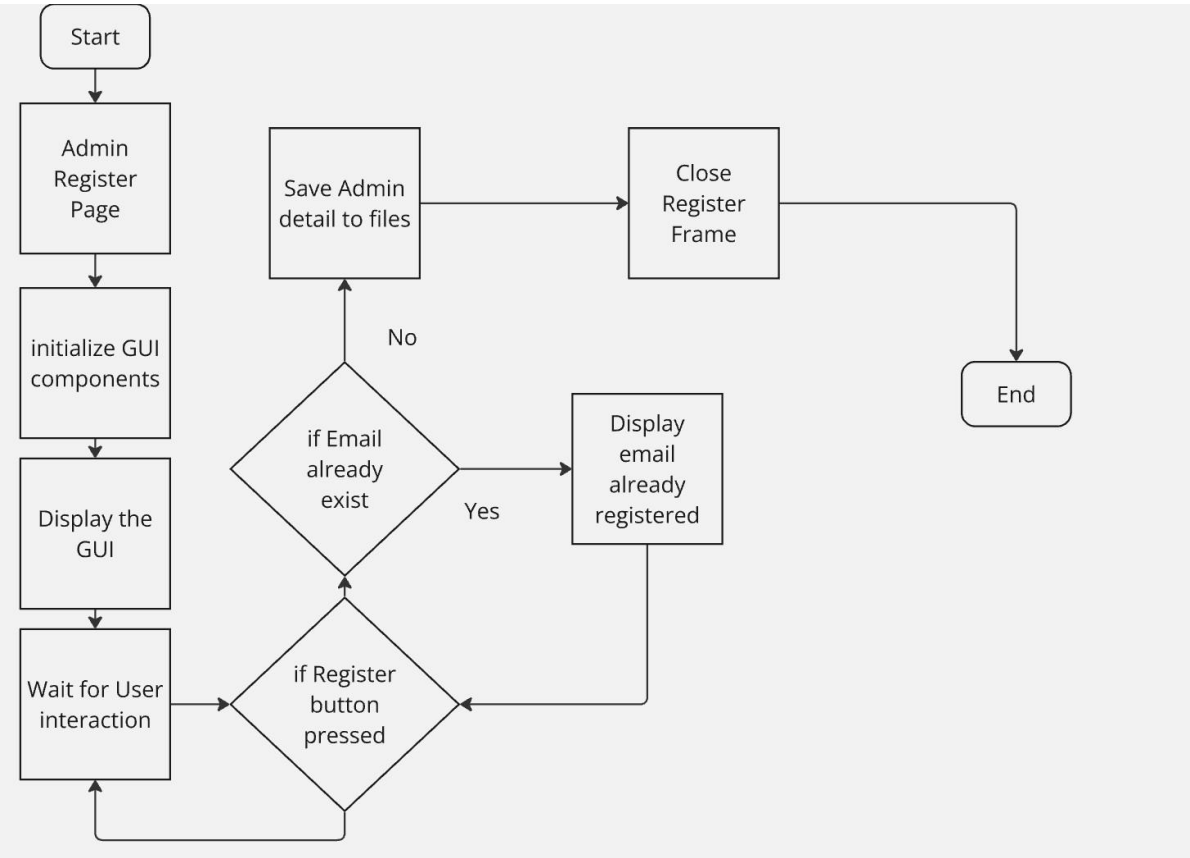


Figure 12.1 Admin Register Page flowchart

+ class AdminRegisterPage - extends JFrame – implements ActionListener
- Emaildocumentpath: String - Passworddocumentpath: String - Register: JLabel - Emaillabel: JLabel - codelabel: JLabel - Registerbutton: JButton - Email: JTextField - PasswordEnter: JPasswordField - textfieldpanel: JPanel - SouthPanel: JPanel - admin: Admin
+ AdminRegisterPage(admin: Admin) + actionPerformed(e: ActionEvent): void

Figure 12.2 AdminRegisterPage class UML Diagram

3.3 Admin Current Week Page:

Current week schedule

Date + week

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Breakfast							-
1st Lunch							
2nd Lunch **(Downstairs)**							-
2nd Lunch **(Stair)**							-
Supper							

Figure 13.0 Admin Current week page interface

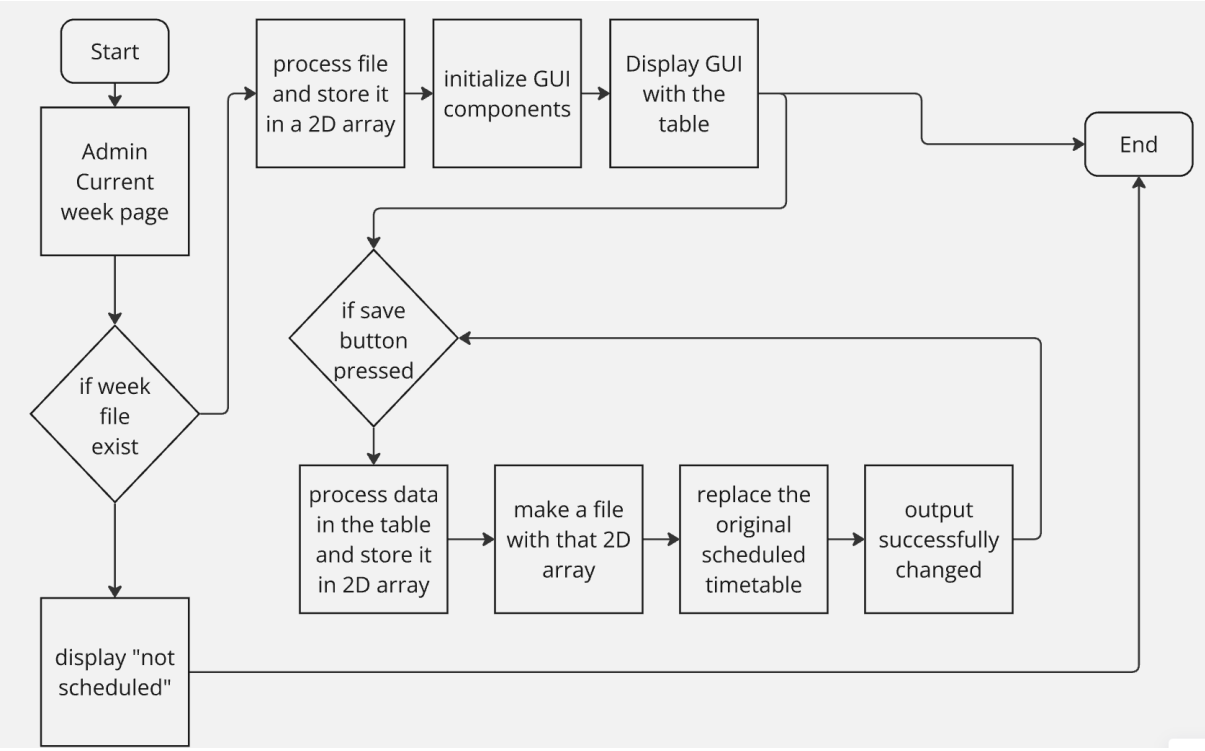


Figure 13.1 Admin Current Week Page flowchart

+ class AdminCurrentWeekPage - extends JFrame – implements ActionListener
- titlelabel: JLabel - table: JTable - rowHeaderTable: JTable - tablemodel: DefaultTableModel - saveButton: JButton - date: String - timeslots: String[][] - admin: Admin
+ AdminCurrentweekpage(date: String, admin: Admin) + actionPerformed(e: ActionEvent): void - processFile(file: File): String[][]

Figure 13.2 AdminCurrentWeekPage class UML Diagram

4. Design of output HTML file:

The output should be clear on the schedule week data with week schedule and table. It should also be able to display start of week date.
The system should allow admin to save it locally with a file explorer.

Schedule week 2025-02-24

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Breakfast	X	e	X	c	X	X	X
1st Lunch	b	X	d	e	b	X	e
2nd Lunch (Downstairs)	b	b	X	X	X	d	X
2nd Lunch (Stairs)	X	c	c	a	c	d	X
Supper	d	a	X	e	X	X	e

5. Design of saved registered data:

In the registered data it will be separated with a character like "|" to separate each user/admin's password and data. There will be 4 different file, User's username, password and Admin's username and password.

```
abc|
abcdefg|
123|
abcde|
admin|
admin1|
```

Test plan: (refer to Criterion A – success criterion)

Test number	Test action	Test result
1	Verify that after opening the application, both User and Admin is directed to the Welcome page, with all JFrame component displayed at specific position	After opening the application, the test action satisfied all users.
2	Verify that after clicking the ForgetPassword button on Welcome frame, User register page and component are accurately displayed	After clicking on the Register button on the Welcome frame, the test action was satisfied.
3	Verify that after clicking on Change password button on Welcome frame, change password page and component are accurately displayed	After clicking on the Change password on the Welcome frame, the test action was satisfied
4	Verify logic of button and ActionListener and extract from textfield and passwordfield data is accurate and successfully stored in password and email file. Ensure it works both in Register and change password page.	All English characters are allowed and successfully stored within the text file
5	Verify that login button in Welcome frame works as it can verify the email and password accurately and perform successful login	All test action was satisfied after login.
6	Verify that Main Frame for User is accurately presented with all JFrame components with working buttons	After a successful login for user, it satisfies the test action.
7	Verify that in User Main Frame when View week button clicked the choose week view page for User is displayed.	Button worked successfully with the Choose week view page.

8	Verify the Usercurrentweekpage aligns with the design and table created smoothly. Test if the data is implemented and displayed correctly and if table aligns with the design	All interface component for User week page display smoothly. All data are displayed with some layout error. But repaired after testing.
9	Verify that Choose week page for user scheduling display smoothly and accurately with accurate data collection and calculation.	Week number should be initialized before the program. All interface work well apart from buttons for each week
10	Verify that the layout and function of the button works well and align with the design.	By setting up 2 extra classes button appear and functions perfectly
11	Verify that the scheduling table and interface for User matches with the success criteria, User should be able to click on cell and turn it into available or unavailable	Sometimes when user click on the cell it might not be working properly. Therefore, the solution is to add another class to prevent error.
12	Verify that scheduling page works well and the save button accurately save the file into specific directory with correct data of each User.	All function performs accurately with efficient and user-friendly interface
13	Verify that if an Admin login it redirects them into an Admin Main frame which is more powerful and more functionality.	Admin Main Frame display with 4 different buttons representing different Admin operations.
14	Verify that Admin can successfully create another admin to have access to same number of functions. It should also save login detail onto another file that are separate from User login detail files.	Admin can create another admin on another fully functioning register page for Admin
15	Verify that in Admin view week page layout and display work the same as User. Admin also need to be able to edit the table for any change. Ensure save button works perfectly when an Admin wants to store changed data	Admin can simply edit the table by clicking on the interface which is user-friendly. It automatically changes the schedule and immediately after change user can access new data.

16	Verify that Admin can save scheduled data as a html file externally onto local device	Admin can click on the output schedule button on the mainframe to save file.
17	Verify that Admin can store have a file chooser so Admin can save wherever on the device as they wish	File chooser works properly and efficienty
18	Verify that Scheduling algorithm aligns with success criteria with a maximum time efficiency O(N^3)	Algorithm works perfectly with most success criteria achieved.
19	Verify that in Admin Main Frame when schedule current week button is clicked scheduling algorithm successfully takes in all user files and	Button and functions work well with little bit of algorithmic error to fix.
	perform scheduling. After scheduling it should store the result in anther txt file as final	Successfully stored within specific directory.
20	Verify that the whole system matches with design and all component work as well as Frame and directory.	All testing finished. Everything works correctly and accurately.